Automated Palo Alto firewall policies conversion to Versa Concerto v12 SASE policies

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Version: 1.1, 19/05/2026
Release notes: Added support for Palo Alto native XML config file. Just copy PAN “sp-config.xml” file into the main working directory. The scripts will detect the existence of that file and will automatically convert. I also created a script to delete ALL objects in Concerto. It’s very useful in a test environment to recycle a tenant for testing. The script will automatically delete all objects from Concerto without the need for you to re-create a new tenant. Refer to section “Tip and precautions” in this document.

Purpose: As the title of this document says, this document describes what the set of scripts do and can’t do. The scripts is capable of cleaning, extracting, modifying and converting Palo Alto firewall policies and its dependent objects and push all configuration to Versa Concerto SASE policies (this is NOT intended for Concerto SDWAN policies). The whole process is fully automated, except when there are unresolved objects and/or configuration that must be manually addressed. In such case, you’ll be given the chance address the issues before the scripts do the conversion.

What is supported:

- PAN policies that use ip-address objects, nested objects groups, predefined/custom services, predefined applications, custom URL categories, security URLF profiles, default policies security profiles, LDAP DN users/groups, convert LDAP users/groups to SCIM
- Different types of objects: static ip-address, ip-range or fqdn-objects
- Convert only objects that are being used in policies
- Import custom services objects, recognize which objects are being used in policies
- Convert PAN predefined services references to Versa predefined services
- Convert PAN predefined applications references to Versa predefined applications
- Convert PAN custom URL categories objects to Versa format
- Convert PAN URLF profiles configuration to Versa format
- Recognize if policies are using security profiles and configure the Versa equivalent

Before you execute the conversion:
When you untar the file, there is a sample PAN configuration that you can use to run the conversion. The sample configuration file gives you an idea of the configuration type that the scripts can work with.

Pre-requisites:
Before you execute the conversion process, these are the pre-requisites that the scripts need during conversion process:

- Export PAN configuration in text file. Make sure you export not only the firewall policies, but also all dependent objects. Eg, ip-objects, profiles, groups etc… Aggregate everything in a single file named “source-pan-rules.txt”, the scripts are designed to correctly recognize the different configuration and extract what they need for the conversion
- If the policies are using LDAP users/groups, you must export all LDAP users and groups from Concerto before executing the conversion. The requirement is that you group DN objects into 2 separate sections. One section for users and the other for groups. The section must be separated by these delimiters (without the quotes): “----BEGIN USERS----”, “----END USERS----“, “----BEGIN GROUPS----” and “----END GROUPS----“. Save the file as “ldap-source-input.txt. There’s a sample LDAP file for your reference. You can extract the LDAP users and groups from Concerto by following the steps documented in this document.
- Review the file: “../miscellaneous/PAN-to-Versa-applications.txt”. This file list some often-used predefined applications that can be used by the scripts to convert PAN predefined applications to Versa. Add more applications in this file if needed.
- Review the file: “../miscellaneous/PAN-to-Versa-urlf-categories.txt”. This file list all the current Versa URL categories as of writing of this document. It’s a one-one conversion from PAN to Versa
- Review the file: “../miscellaneous/PAN-to-Versa-services.txt”. This file list some often-used predefined services that be used by the scripts to convert PAN predefined applications to Versa. Add more services in this file if needed.
- The scripts must be run on MacOS or Linux with Python v3.9 or later.

Instruction:
To start the conversion process, run the wrapper script “./RunMe.sh”. The wrapper script will automatically run all necessary scripts located in sub-directory “../scripts”. As stated before, the whole process is automated, except for when there are unresolvable elements that need your intervention. Those exceptions will be described below.

The whole conversion process framework is designed to be modular and divided into many steps. Some steps will require your input and some steps will output information for your review.

The scripts are designed to be automated with minimal user intervention during the process. However, there are 2 steps that require user intervention. Once before the process begins and a hard pause before the actual conversion and POST to Concerto begins. Both steps are explained below.

Beginning the process:
At the beginning of the process, and after you enter specific Concerto related information, the script will ask you about LDAP profile and SCIM profile. The script can do 2 things: 1-convert the original policies with LDAP to Concerto policies with LDAP, or 2-convert the original policies with LDAP to Concerto policies with SCIM.

- **From LDAP to Concerto LDAP**: enter the LDAP profile as prompted. Press ENTER at the SCIM profile prompt. Then the script will ask you which LDAP format is on you Concerto setup. There are 3 different formats on Concerto, depending on how AD server was configured. You must choose the correct format, else the policies POST will fail. You can determine the format by simply doing this: go onto Concerto GUI, choose to add an “Internet Protection Rule”, go into the section “Users & Groups”, click on “Customize” users and groups, select “Selected Users” option,

choose the correct LDAP profile from the drop-down menu. Check both formats for groups and users. First, determine the groups format. If your groups use DN format with UID trailing (CN=group,DC=domain,DC=com|S-1-2-3-547683-23424), then you must choose option #3. If that's not the case, verify your users format. It's either UPN (user@domain.com) or DN (CN=user,DC=domain,DC=com), if that's the case then choose either option #1 or #2.
- **From LDAP to Concerto SCIM**: enter the LDAP profile as prompted and enter the SCIM profile name. The scripts will automatically fetch the correct information, there's nothing else you need to do. NOTE: the script will fetch all SCIM users and groups, this may take a while if you have a massive list of users and groups. You still need to provide a called "ldap-source-input.txt". Follow the instructions below to extract LDAP users/groups from Concerto.

Pre-conversion steps:
After all the process of preparing the original configuration for conversion to Concerto, the script will pause and wait for you to press ENTER. After which, the scripts will do the actual configuration conversion and POST them to Concerto. However, it's most likely that you need to provide additional information for the scripts to do the conversion. Verify the existence of these files and provide appropriate inputs.
- Zone-conversion.txt: the file lists all the zones used in the original configuration, provide the equivalent in Concerto. NOTE: Concerto provides 2 default zones, "ptvi" (SDWAN) and "remote-client" (SASE VPN).
- Bonus script: there's a script "../scripts/remove-all-objects.py". That script will help you remove all objects from Concerto for a particular tenant.

How to extract LDAP objects from Concerto manually:
If you're converting policies with LDAP, you'll need to provide the scripts with a list of LDAP users and groups in a text file format like described earlier in this document. This helps the script in the early stage of verification to distinguish between users and groups objects contained in the original configuration. There are 3 scripts to help you do extract LDAP objects from Concerto. They're located in the subdirectory "scripts".

- Get-token.py: this script will get the authentication and bearer tokens need to talk to Concerto
- Get-ldap-users.py: this script will get the users
- Get-ldap-groups.py: this script will get the groups

Run all 3 scripts in that order. Aggregate the users and groups files into one file "ldap-source-input.txt" with the delimiters as explained earlier.

Tip and precaution:
- Very important to pre-populate the files in subdirectory "miscellaneous". The scripts need to know the Versa equivalent of PAN predefined services/application/URL to convert to. If you don't know which PAN services/application/URL are being used in your policies, execute the scripts as a dry run. You will see a file that will list all PAN predefined services/application/URL that are being used.
- There's a script in the the sub-directory "../scripts/remove-all-objects.py". Run the script to delete all objects in a tenant. Run the script with the option "--quick" and it will send a DEL command every 0.5s. It's faster because the script won't wait for Concerto confirmation after every DEL

command. However, if you have thousands of objects, it can overwhelm Concerto and the server will stop executing DEL command. In such case, run the script without the “quick” option.

Description:

- **Step0**: this step will clean the input file “source-pan-rules.txt” for any undesirable characters that will affect the conversion process later. It will list all firewall policies that use security profiles, this is informational. The script will convert these security profiles configuration to Versa default. Lastly, it will output the file “zone-conversion.txt”. YOU MUST manually type in the corresponding zones on Versa platform.
- **Step1**: this step will clean and convert all the policies names to Versa supported format. It will prompt you for the maximum length that you wish to use, default is 63 characters. The script will truncate any policy name to that length. It will also clean and replace any unsupported characters with “_”. If you wish to review which policies were affected, there’s an output file “../step-1/ step-1_modified-policy-name.txt”
- **Step2**: this step mainly focuses on ip-objects and groups of objects extraction and sanitizing. It will extract all those objects, truncate the objects names to maximum 63 characters, replace any unsupported characters with “_”. It will automatically add “/32” to any ip-address object that doesn’t have the “/32” subnet mask. Lastly, it will output 2 filles that list all objects that the script can’t resolve, “must-fix-address-object.txt” and “must-fix-address-group-object”. YOU MUST review and resolve those objects later on before the conversion.
- **Step3**: this step will correlate all ip-address objects and objects groups with the actual firewall policies. It will remove all objects that are not used by the policies or any ip-address objects that are not used in any objects groups. The script is able to recognize and correlate nested groups, that is groups within groups. Note: this step could take a while if you have thousands of objects.
- **Step4**: this step mainly focuses on custom service objects. It will extract and sanitize the objects names. It will truncate any object name that’s longer than 31 characters. It will replace all unsupported characters with “_”. It will correlate with the actual firewall policies and change the reference name in the policies. The script will output any service object configuration that the script can’t resolve or recognize. Review the file “../step-4/ step-4_unsupported-service-config.txt”
- **Step5**: since Versa do not support services groups, the script will extrapolate the group service members and replace all firewall policies refence with the corresponding individual service object names. This step will also extract predefined service and application objects and it will convert PAN predefined service and application to Versa’s version based on the files “../miscellaneous/ PAN-to-Versa-applications.txt” and “../miscellaneous/ PAN-to-Versa-services.txt”. If there are no equivalent service or application in the conversion file, the script will use whatever object name that’s currently in the PAN configuration. **Note: those files contain some of the mostly used services and applications, it’s not complete. YOU MUST, edit the files and insert the Versa equivalent. Else, the import of the service template will fail if you’re not using the correct service or application name.**
- **Step6**: this step will extract custom URL categories configuration. It will sanitize the object names, truncate the length and replace unsupported characters with “_”. The script will also rectify any modified names in the firewall policies reference.
- **Step7**: this step will extract all LDAP users and groups from the firewall policies and it will divide the LDAP objects into separate users and groups based on the reference file “ldap-source-input.txt”. The script is able to recognize LDAP DN, UPN and canonical formats and it will process the objects accordingly.

- **Step8**: this step will sanitize and convert the LDAP configuration lines to Versa format. It will also convert any non-DN users/groups format in the policies to DN format.
- **Step9**: this step will convert and transfer all PAN address objects to Versa service template. It will also do a final cleanup and CIDR formatting of any objects with the wrong subnet mask.
- **Step10**: this step will convert and transfer all PAN groups objects to Versa service template.
- **Step11**: this step will convert and transfer all PAN custom services objects to Versa service template.
- **Step12**: this step will convert and transfer all PAN custom URL categories objects to Versa service template.
- **Step13**: this step will convert and transfer all PAN security URLF profiles to Versa service template. This will also output any unsupported predefined URL categories in the file “unknown-urlf-category.txt”
- **Step14**: this is the last step where the script will transfer and convert PAN firewall policies to Versa template.